AdaDINO-LA: A Controlled Study of DINOv2 Adaptation Strategies for Left Atrial Segmentation in LGE-MRI

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Abstract

Left atrial (LA) segmentation from late gadolinium-enhanced MRI (LGE-MRI) supports atrial fibrillation studies, but manual delineation is slow and observer-dependent. Deep networks automate it but need many annotated scans; pretrained foundation encoders such as DINOv2 may reduce this need. Such an encoder can be frozen, partially or fully fine-tuned, or adapted with low-rank adaptation (LoRA), but to our knowledge no previous LA study compared these strategies under one protocol. We compared the four strategies on DINOv2 ViT-Large and three on ViT-Giant, with the same decoder, loss, model-selection rule, three seeds, and 154-volume split. Against the same ViT trained from scratch without augmentation, DINOv2 initialization raised the Dice coefficient from 0.764 to 0.907. Adaptation strategy changed Dice by only 1 to 3 percentage points, and ViT-Giant, at 2.6 to 3.3 times the memory, added at most 0.6; a skip-connection decoder added at most 0.5 with all training volumes and lowered Dice with few. With one improved recipe applied post hoc to all ViT-Large strategies, LoRA and full fine-tuning did not differ detectably (both 0.912), with LoRA using 56% of the peak training memory but not less training time; in post hoc runs with this recipe, the gain over the ViT from scratch was 15 points with all 108 training volumes and 42 with 11, and 21 and 37 when both were trained with nnU-Net-style augmentation. The recipe thus changed the strategy ranking. However, a task-specific 3D nnU-Net remained more accurate in-domain, even against augmented LoRA, and on external bSSFP MRI.

Keywords: Left atrial segmentation, LGE-MRI, DINOv2, Fine-tuning, Low-rank adaptation, Computational efficiency

1. Introduction

Accurate delineation of the left atrium (LA) supports quantitative analysis of cardiac magnetic resonance images in atrial fibrillation studies, including scans acquired before and after catheter ablation [1]. In late gadolinium-enhanced magnetic resonance imaging (LGE-MRI), atrial anatomy must be distinguished from surrounding tissue despite variable contrast and complex boundaries. Manual annotation is labor-intensive and subject to intra- and inter-observer variability, motivating automated segmentation [1, 2]. We address the binary LA endocardial segmentation target supplied with the dataset, not scar classification or outcome prediction.

Task-specific frameworks such as nnU-Net, configured and trained from scratch for each dataset, remain strong references for medical segmentation [3–5]. Because they learn all features from the

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annotated scans of the target dataset, each new dataset or acquisition setting typically requires its own labeled training set. Self-supervised foundation models, notably DINOv2, instead learn patch-level features once from large unlabeled natural-image collections and reuse them for dense prediction [6], which may reduce the annotated scans needed [7] and improve robustness across scanners and protocols [8]. For LA segmentation, DINOv2 encoders have been reported to remain accurate with limited training data [9] and, with U-Net-style decoders, to exceed a 2D nnU-Net [10]. A practitioner adopting a DINOv2 encoder must decide whether to keep it frozen, partially fine-tune it, adapt it with low-rank adaptation (LoRA) [11], or fully fine-tune it; whether a larger backbone is worth its cost; and how much of the accuracy is owed to pretraining, particularly when few scans are annotated.

Existing evidence does not resolve these decisions for LA segmentation. The closest studies, by Kundu et al., evaluated DINOv2 across backbone sizes [9], paired it with different decoders [10], and added multi-scale feature fusion [12], but with different decoders, partitions, and reference pipelines. Broader benchmarks compared fine-tuning strategies for SAM [13], parameter-efficient adaptation for few-shot volumetric segmentation [14], and transfer paradigms for 3D tasks [15], but not for LA segmentation in LGE-MRI, and the benefit of natural-image pretraining cannot be assumed for medical tasks [16]. Resolving these decisions requires one protocol that varies adaptation strategy, backbone scale, and pretraining while holding other factors fixed, because method rankings can change with evaluation choices [17], and requires training resources to be measured directly.

We therefore conduct a controlled empirical comparison, termed AdaDINO-LA, in which the principal experimental design factors are held fixed where possible. AdaDINO-LA denotes the study design, not a new network, and control here means a shared protocol with nominally fixed hyperparameters and no augmentation, rather than per-strategy tuning. The study addresses four research questions:

RQ1. How much does DINOv2 initialization contribute compared with the same architecture trained from random initialization, and how does this contribution change when fewer annotated training volumes are available?

RQ2. Under the shared protocol, how do adaptation strategy and backbone scale affect accuracy and training resources?

RQ3. How sensitive are these effects to the training recipe, namely the adapter learning rate, gradient clipping, the decoder readout, and decoder capacity, and how do the strategies compare when all of them receive one common improved recipe?

RQ4. How do the adapted models perform on external data from other centers and from another MRI sequence?

The study makes three empirical contributions. First, it compares four adaptation strategies and two backbone sizes under one split, preprocessing recipe, decoder topology, loss, and checkpoint rule, with three seeds, paired case-level statistics, and a no-pretraining control. Second, it measures training memory, time per batch, and training time. Third, it tests which conclusions persist under one common improved recipe, a larger decoder with skip connections, reduced label budgets, and external data. Throughout, a strong task-specific reference, the residual-encoder 3D nnU-Net, is trained and scored on the same cases to place the adapted models relative to an established pipeline. The findings are condensed into practical recommendations.

2. Related work

2.1. Left atrial segmentation and task-specific pipelines

U-Net introduced a widely adopted encoder–decoder design for biomedical segmentation, and nnU-Net showed the importance of configuring preprocessing, architecture, and training together [3, 4], with rigorous validation against strong baselines [5]. UNETR connects a transformer encoder to a convolutional decoder through multi-resolution skip connections [18]. LA analysis from LGE-MRI includes cavity, wall, and scar segmentation [1]; task-specific LA networks include the compact 3D Usformer [19] and the semi-supervised CA-Net [20], and a multicentre study evaluated 2D and 3D bi-atrial LGE-MRI segmentation [21]. Scores from other datasets, partitions, or targets are not treated as comparable here.

2.2. Vision foundation models in medical imaging

DINOv2 provides patch tokens that can be reshaped into spatial feature maps [6], which makes frozen extraction a natural baseline for dense prediction. DINOv2 has been evaluated on radiology benchmarks [7] and, with other foundation models, for domain generalization in medical segmentation [8]. Self-supervised pretraining has also been studied directly on 3D medical images, for example for a Swin UNETR encoder pretrained on CT volumes [22].

Kundu et al. evaluated DINOv2 for LA segmentation across backbone sizes and limited-data settings on LAScarQS 2022 [9]. A second study paired DINOv2, SAM, and MedSAM encoders with linear or U-Net-style decoders on LAScarQS 2022 and on the 2018 challenge data used here; with the U-Net-style decoder, DINOv2 exceeded a 2D nnU-Net on both, whereas with linear decoders the 2D nnU-Net was better on the 2018 data [10]. A third used a frozen DINOv2 encoder with learned multi-scale fusion and image information in the decoding pathway and reported higher Dice than a 2D nnU-Net baseline [12]. Outside DINOv2, a ViT pretrained as a masked autoencoder on the target LGE-MRI data has been fine-tuned with a single-scale adapter for LA segmentation [23].

Beyond DINOv2, MedSAM adapts a promptable segmentation model across modalities [24], a large-scale evaluation documents the limits of direct SAM transfer to medical images [25], a specialization study compares foundation-model pipelines with nnU-Net [26], and a one-shot study builds its segmentation pipeline from foundation models [27]; their supervision regimes and targets differ from ours.

2.3. Parameter-efficient adaptation

LoRA represents a weight update as the product of two low-rank matrices while retaining the pretrained weights [11]; partial fine-tuning instead updates a subset of the original layers. Parameter-efficient adaptation was first benchmarked in medical imaging for classification and generation [28]; in segmentation, SAMed trains LoRA updates in the SAM image encoder [29], Medical SAM Adapter and Trans-SAM insert adapter modules [30, 31], HF-SAM combines LoRA with multi-scale fusion [32], and LoRA with a DeepLabV3 decoder has adapted DINOv2 for oral lesion segmentation [33]. Because frozen weights still take part in the forward pass and gradients must propagate through the backbone, a small trainable fraction need not imply small memory or short training; a SAM benchmark found that parameter-efficient strategies improved accuracy but not training efficiency [13]. AdaDINO-LA introduces no new adapter (Table 1).

Table 1. Positioning relative to selected prior work. Entries describe research objectives rather than an exhaustive absence/presence audit. Published scores are not compared across protocols.

Study	Main setting	Focus relative to this study
Usformer; CA-Net [19, 20]	Task-specific 3D LA segmentation	Architectural development; CA-Net uses semi-supervised learning.
Kundu et al. [9]	DINOv2 for LA segmentation	Backbone-size and limited-data evaluation on LAScarQS; directly related transfer setting.
Kundu et al. [10]	DINOv2, SAM, and MedSAM encoders for LA segmentation	Linear versus U-Net-style decoders on LAScarQS and the 2018 challenge data.
Kundu et al. [12]	LA segmentation with multi-scale fusion	Learned fusion and image-driven spatial integration in the prediction pathway.
SAMed; medical adapters; HF-SAM [29–32]	SAM-derived medical segmentation	LoRA/adapters and task-specific decoding or fusion mechanisms.
Adaptation benchmarks [13–15]	Fine-tuning strategies for foundation models across organs and modalities	Strategy rankings across backbones and decoders; not LA LGE-MRI.
AdaDINO-LA	DINOv2 for LGE-MRI LA segmentation	Common decoder topology and evaluation; four adaptation strategies and two backbone sizes over three seeds; no-pretraining control; recipe sensitivity; external evaluation; multiple resource measures.

3. Method

3.1. Problem formulation and study design

Let X_i denote an LGE-MRI volume and Y_i its binary reference mask. For slice z, preprocessing produces an input $x_{i,z} \in \mathbb{R}^{3\times448\times448}$. An encoder f_θ and decoder g_ϕ predict logits

$$s_{i,z} = g_\phi(f_\theta(x_{i,z})), \qquad p_{i,z} = \mathrm{sigmoid}(s_{i,z}). \tag{1}$$

Thresholded masks are restored to the native in-plane geometry and stacked to obtain the volume prediction $\widehat{Y}_i$. The network processes 2D slices; the evaluation unit is a reconstructed 3D volume.

On ViT-Large we compare frozen extraction, partial fine-tuning, LoRA, and full fine-tuning; on ViT-Giant, frozen extraction, LoRA, and partial fine-tuning of the last four blocks. Giant Full FT was not run because at batch size 4 it was not expected to fit the 24 GB GPUs used for most runs (Giant LoRA alone peaked at 15.9 GB). Names combine backbone and strategy, abbreviating fine-tuning as FT (for example, Large Full FT). A Large frozen configuration trained with binary cross-entropy (BCE) alone evaluates the loss choice, and identifiers E0–E7 in the supplementary material link results to archived runs.

3.2. DINOv2-based segmentation framework

We use pretrained DINOv2 ViT-Large/14 and ViT-Giant/14 encoders [6], with embedding dimensions $C = 1024$ and $C = 1536$. Both use 14×14 patches and produce the same 32×32 grid of $N_{\mathrm{tok}} = 1024$ spatial tokens, so the decoder topology is unchanged (Figure 1). Normalized patch tokens from the final encoder block are reshaped to $C \times 32 \times 32$, and non-spatial tokens are omitted. A 1×1 convolution projects the features to 256 channels, followed by batch normalization and ReLU. Three bilinear upsampling blocks (3×3 convolution, batch normalization,

ReLU; 128, 64, and 32 channels) raise the resolution to 256×256, and a final interpolation to 448×448 is followed by a 16-channel convolutional block and a one-channel output. The decoder has no encoder skip connections, and only its input projection depends on C.

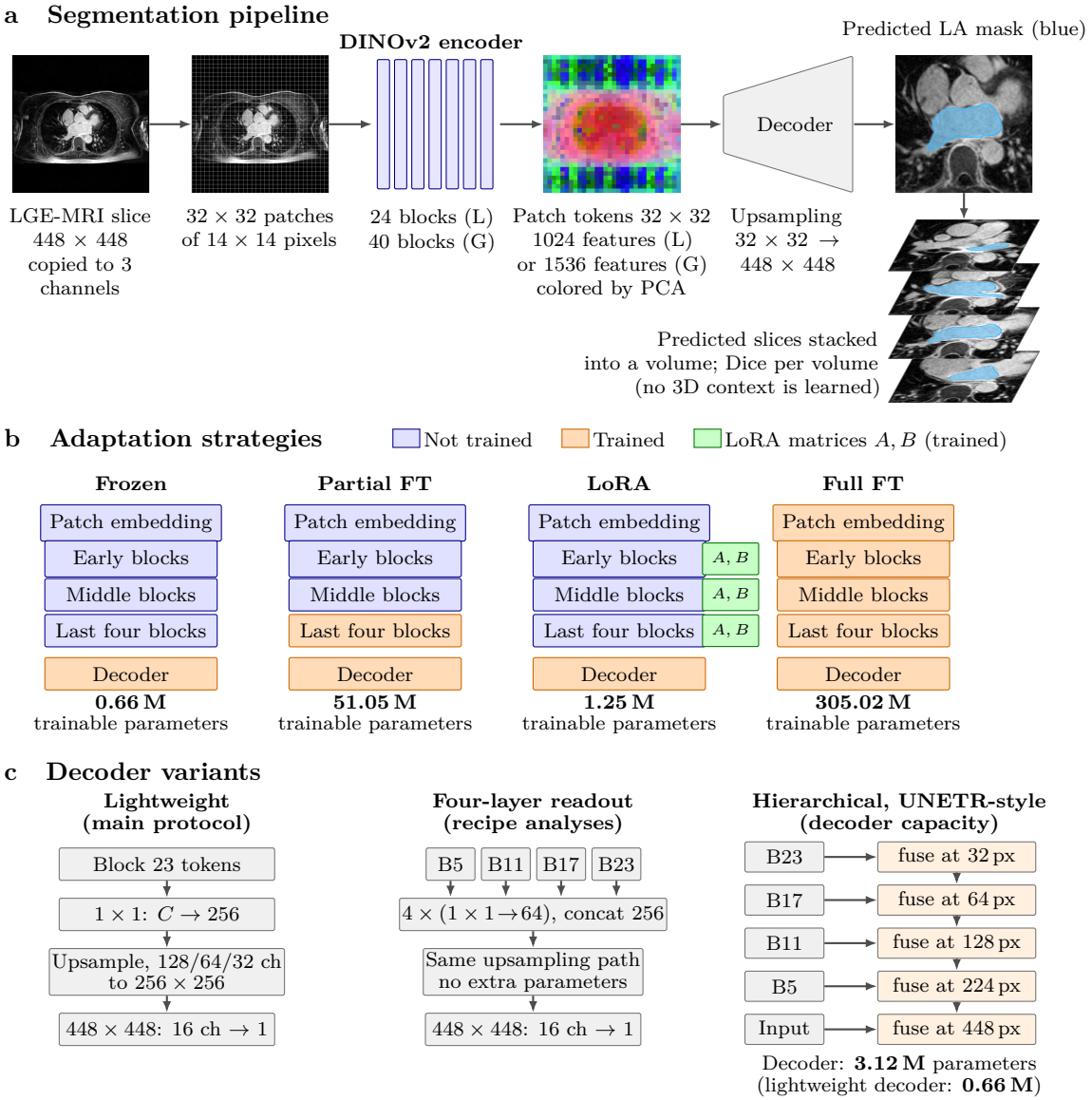


Figure 1. DINOv2 segmentation framework. (a) Each preprocessed slice is split into 14×14 patches; the DINOv2 encoder yields a 32×32 grid of patch tokens (shown as the first three principal components of pretrained ViT-Large tokens), which the decoder upsamples to a slice mask; slice masks are restored to the native geometry and stacked, so evaluation is volumetric but no 3D context is learned. Images: median-ranked test volume, seed-42 Large LoRA prediction. (b) Parameter groups that are fixed, directly updated, or adapted with LoRA ($W' = W + \frac{\alpha}{r} BA$ in all attention projections), with trainable parameter counts for ViT-Large; all four strategies are evaluated on ViT-Large, and Frozen, Partial FT, and LoRA on ViT-Giant (last four blocks: 20–23 for Large, 36–39 for Giant). (c) The lightweight decoder of the main protocol, the four-layer readout of the recipe analyses (tokens of blocks 5, 11, 17, and 23), and the hierarchical decoder of the decoder-capacity analysis (Section 5.7; details in the supplementary material). The recommended starting point (Table 9) combines LoRA in (b) with the four-layer readout in (c).

3.3. Adaptation strategies

Frozen. All encoder parameters are frozen and the encoder runs in evaluation mode; only the decoder is optimized.

Partial fine-tuning. The last four Transformer blocks (zero-based indices 20–23 for Large and 36–39 for Giant) are updated together with the decoder; the patch embedding, earlier blocks, and final normalization remain frozen. Four blocks are 4/24 of Large depth but 4/40 of Giant depth, so the comparison matches block count, not depth fraction or trainable parameter count.

Full fine-tuning. All ViT-Large encoder parameters and the decoder are trainable, with a smaller learning rate for the encoder than for the newly initialized decoder.

LoRA. For a pretrained linear weight $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the effective weight is

$$W' = W + \frac{\alpha}{r} BA, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad B \in \mathbb{R}^{d_{\text{out}} \times r}, \tag{2}$$

with W frozen. Adapters are injected into the joint query–key–value projection and the attention output projection of every block; A uses Kaiming uniform initialization and B zeros, so the initial update is zero. Both backbones use the same rank and scaling (Table 2).

3.4. Training objective

For a batch of B slices with N pixels each, we optimize

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{BN} \sum_{k=1}^{B} \sum_{j=1}^{N} \left[y_{kj} \log p_{kj} + (1 - y_{kj}) \log(1 - p_{kj}) \right], \tag{3}$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{B} \sum_{k=1}^{B} \frac{2 \sum_j p_{kj} y_{kj} + \epsilon}{\sum_j p_{kj} + \sum_j y_{kj} + \epsilon}, \tag{4}$$

$$\mathcal{L} = \tfrac{1}{2} \mathcal{L}_{\text{BCE}} + \tfrac{1}{2} \mathcal{L}_{\text{Dice}}, \qquad \epsilon = 10^{-6}. \tag{5}$$

BCE is computed from logits, and soft Dice per slice, averaged over the batch. The BCE-only control both doubles the BCE weight and removes Dice, so it measures objective sensitivity for frozen features rather than the effect of Dice alone; the shared objective is not claimed to be optimal for each strategy.

4. Experimental setup

4.1. Dataset and fixed case-level split

We study 154 LGE-MRI volumes of the local Utah2018 collection [2], each with an LA endocardial reference mask, split into 108 training, 15 validation, and 31 test volumes (split seed 2024; not the official benchmark split). All 88 slices of each volume are retained, giving 9,504, 1,320, and 2,728 slices.

The public collection includes pre- and post-ablation scans [34] but no reliable volume-to-patient mapping, so the split is by volume and a case denotes a volume. An image-similarity audit of all 11,781 volume pairs identified one train–test pair with identical masks (supplementary material); excluding that test volume is a sensitivity analysis (Supplementary Table S3).

4.2. Preprocessing and training

Intensities are clipped at the 0.5th and 99.5th percentiles of each volume (finite nonzero values when zero background is present), min–max normalized, and stored as 8-bit slices. Each slice is zero-padded to a square and resized to 448×448 (bilinear for images, nearest-neighbor for masks), replicated across three channels, and normalized with ImageNet statistics. No augmentation is used except in one post hoc analysis (Section 4.4), so the comparisons concern adaptation under a fixed input distribution; this defines the protocol rather than asserting that augmentation would not help.

Each DINOv2 configuration is trained with seeds 42, 2026, and 3407 under the settings of Table 2, with a newly initialized decoder in each run, mixed precision with dynamic gradient scaling, and the loss computed in FP32 (supplementary material). At each epoch, the checkpoint with the highest mean 3D Dice over the 15 validation volumes at native resolution is kept, and training stops after ten epochs without improvement; test metrics are never used for selection.

Experiments ran on four GPU models with PyTorch 2.8.0 or 2.11.0 (Table S7). Retraining the same configuration and seed on another GPU model changed Dice by 0.38 to 0.60 points in three pairs, comparable to seed-to-seed variation, so differences of this size between runs on different GPU models are not interpreted as effects. The main Large Full FT runs also differ from the other main-table rows in GPU, PyTorch version, and DINOv2 source snapshot.

Table 2. DINOv2 training settings of the main protocol. Backbone learning rate applies only to unfrozen parameters or LoRA adapters. Settings varied in the recipe-sensitivity analyses are described in Section 4.4.

Setting	Value
Pretrained encoders	DINOv2 `dinov2_vitl14` and `dinov2_vitg14` (non-register)
Input / patch / grid	448×448 / 14×14 / 32×32
Training seeds	42, 2026, 3407
Optimizer	AdamW; $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$
Learning rates	Decoder: 10^{-3}; backbone/adapters: 10^{-5}; constant schedule
Weight decay	0.01 for weight matrices; 0 for biases, normalization, and other one-dimensional parameters
Batch size	4 slices; gradient accumulation: 1
Partial FT blocks (zero-based)	Large: 20–23; Giant: 36–39; decoder trainable
Training limit	50 epochs; early-stopping patience: 10
Precision	AMP FP16; dynamic GradScaler; loss computed in FP32 for all configurations
Gradient clipping	None in the main protocol
LoRA configuration	$r = 4$, $\alpha = 4$; all attention qkv/proj layers; no LoRA dropout
Model selection	Highest mean validation volume Dice
Prediction	Sigmoid threshold 0.5; no postprocessing

4.3. Task-specific 3D reference

We train the 3D full-resolution nnU-Net with its medium residual-encoder preset (ResEncM; nnunetv2 2.8.1) as a task-specific reference [4, 5] on the same 108 training and 15 validation cases, planned on these 123 cases only, and score it with the same evaluator and seeds. It keeps its own recipe, including Z-score normalization, $32 \times 256 \times 256$ patches, nnU-Net augmentation, deep supervision, and 1,000 epochs of 250 batches (supplementary material). As a slice-wise

counterpart, fixed post hoc together with the decoder study, the ResEncM 2D configuration was trained with the same recipe and seeds. Because these components differ jointly, the comparison is pipeline-level and does not isolate architecture or pretraining.

4.4. Pretraining control, recipe sensitivity, label budget, decoder capacity, and external evaluation

No-pretraining control (RQ1). The encoder and decoder of Large Full FT are trained with the identical protocol and seeds from random initialization, using DINOv2's own initialization scheme. Because the shared encoder learning rate is low for training from scratch, a second recipe uses an encoder learning rate of 10^{-4} with gradient-norm clipping at 1.0, also with three seeds. Frozen, Partial, and LoRA variants are not evaluated without pretraining, because their frozen weights would be random.

Recipe sensitivity (RQ3). Each factor is first varied, with three seeds, for the strategy it concerns: (i) an adapter learning rate of 10^{-4} for Large and Giant LoRA, chosen in a single-seed validation sweep; (ii) gradient-norm clipping at 1.0 for Large Full FT; and (iii) a four-layer readout for Large Frozen and Large LoRA, in which tokens from blocks 5, 11, 17, and 23 are each projected to 64 channels and concatenated without adding trainable parameters (Figure 1c). (iv) A common improved recipe applies all three to the four ViT-Large strategies, with pretrained weights at 10^{-5} and new parameters at a higher rate (decoder 10^{-3}, LoRA adapters 10^{-4}); its 12 runs used one RTX 5090, and their test set was scored only after all had finished. All recipe analyses were defined after the main test results were known and select on validation data only.

Label budget (RQ1). The common improved recipe is retrained on nested subsets of 11 and 27 training volumes (10% and 25%), drawn once per seed and shared by all configurations of that seed: the four ViT-Large strategies, nnU-Net, and a randomly initialized ViT-Large (full fine-tuning, encoder learning rate 10^{-4}), which is also retrained on all 108 volumes. A reduced-data epoch draws the 2,376 batches of a full-data epoch, so the update schedule and early stopping are unchanged, and no setting was tuned per budget. Subsets and runs were fixed after the main and common-recipe test results but before any reduced-data run (supplementary material).

Augmentation (RQ1). Because the DINOv2 models and the no-pretraining control were trained without augmentation, LoRA and the randomly initialized ViT-Large of the label-budget analysis were retrained on 108 and 11 volumes with nnU-Net-style 2D augmentation of the training slices (rotation, scaling, noise, blur, brightness, contrast, low resolution, gamma, and mirroring at the default nnU-Net probabilities), everything else unchanged (12 runs, one RTX 5090). The design and nine contrasts were fixed after all other results but before any augmented run.

Decoder capacity (RQ3). A hierarchical decoder in the style of UNETR [18], with skip connections from four encoder blocks and the input slice (3.12 instead of 0.66 million parameters; Figure 1c), replaces the lightweight decoder of the common recipe for the four ViT-Large strategies on 108 volumes and for LoRA and Full FT on 11 (18 runs, one RTX 5090). Its design and planned contrasts were fixed after the common-recipe and label-budget results but before these runs.

External evaluation (RQ4). All main-protocol checkpoints and nnU-Net are applied without retraining to two public datasets, and the common-recipe checkpoints are scored post hoc. CDEMRIS provides pre- and post-ablation LGE volumes of 30 patients [35]; 10 patients whose volumes carry Utah-style headers were excluded a priori to avoid overlap with the training data (supplementary material), leaving 40 volumes from 20 patients at two non-Utah centers. Task02 of the Medical Segmentation Decathlon (MSD) provides 20 balanced steady-state free precession (bSSFP) volumes with public masks, a different MRI sequence [36, 37], used after correcting their identity affines. External Dice values are not compared with internal ones because label definitions may differ between centers.

4.5. Metrics and statistical analysis

Slice predictions are thresholded at 0.5, resized to the padded native geometry with nearest-neighbor interpolation, unpadded, and stacked. For binary volumes P and G, Dice $= 2|P \cap G|/(|P| + |G|)$ is the primary outcome and IoU $= |P \cap G|/|P \cup G|$ is complementary; both are one if both volumes are empty and zero if only one is. Metrics are computed per volume, averaged over test volumes for each seed, and summarized by the mean and sample standard deviation (SD) of the three seed means, so the SD describes retraining variation on one split. Dice differences are reported in percentage points (Dice $\times$ 100), abbreviated as points.

For paired comparisons, Dice is averaged over seeds within each case, and the 31 test cases (CDEMRIS: 20 patients with both scans kept together; MSD: 20 volumes) are resampled 20,000 times with all configurations kept paired (seed 20260926). Percentile intervals give 95% confidence intervals (CIs); two-sided bootstrap p values are twice the smaller tail proportion, reported as $p_{\text{Holm}} < 0.002$ when no draw crosses zero. Four families are Holm-adjusted (Supplementary Table S12): the 16 contrasts of Table 4, the 15 label-budget contrasts, the 12 decoder contrasts of Table 6, and the nine augmentation contrasts; other contrasts carry unadjusted intervals, and all tests are exploratory. Because seeds are averaged first, intervals are conditional on the trained models; a contrast is called robust only when its interval excludes zero and all three per-seed differences share its sign (Supplementary Table S2), a screen that a null effect passes by chance in one of four cases. Key contrasts are re-estimated with a case $\times$ seed bootstrap and on the test volumes least similar to the training data (Table S4). Intervals treat test volumes as exchangeable, and an interval containing zero does not establish equivalence.

Writing $D_{b,s}$ for mean Dice with backbone b and strategy s, we report adaptation effects $D_{b,s} - D_{b,\text{Frozen}}$, scale effects $D_{\text{Giant},s} - D_{\text{Large},s}$, and scale interactions $I_s = \Delta_{\text{scale}}(s) - \Delta_{\text{scale}}(\text{Frozen})$ for $s \in \{\text{LoRA}, \text{Partial}\}$, all formed within cases before resampling; changes between label budgets are formed in the same way. Training stability is summarized post hoc by the maximum drawdown of the validation trajectory, MDD $= \max_{u \le t}(D_u - D_t)$, which captures declines even if the run later recovers.

4.6. Training resources

We report trainable parameters and measure peak allocated training memory, without gradient checkpointing, and time per training batch for a fixed workload of 300 batches per configuration on one RTX 4090 (supplementary material), together with the epochs each run completed. Wall-clock training times of the main-protocol runs, which used several GPU models, are not compared; for the 12 common-recipe runs on one RTX 5090, training time is measured directly, excluding validation and checkpoint writing. Configurations that no other configuration matches in mean Dice at no greater cost are marked descriptively for memory, time per batch, and parameters; nnU-Net is excluded because its 3D workload is not matched.

5. Results

5.1. Contribution of DINOv2 pretraining

Under the Large Full FT protocol, the ViT-Large encoder and decoder trained from random initialization reached Dice 0.7639 ± 0.0234 (Table 3). DINOv2 initialization was higher by 14.27 points (95% CI [9.52, 20.38], unadjusted) in all 31 test cases, and even the frozen DINOv2 encoder exceeded the randomly initialized encoder trained end to end, by 11.08 points ([6.36, 17.22]). The second from-scratch recipe was lower still (0.7322 ± 0.0084), so the 14.3-point gap uses the stronger from-scratch recipe; Sections 5.5 and 5.6 report the gain with fewer volumes and with augmentation.

Table 3. Native-resolution volume-level test performance under the main protocol (all DINOv2 rows use one FP32 loss implementation). Values are mean ± sample SD across three training seeds. Rows above the first rule use $0.5\mathcal{L}_{\text{BCE}} + 0.5\mathcal{L}_{\text{Dice}}$; bold marks the highest mean within each backbone block, not statistical significance. The controls below use the same split and evaluator: the BCE-only frozen objective, and ViT-L trained from random initialization with the Large Full FT protocol (no DINOv2 weights). nnU-Net uses Dice+CE and a distinct 3D recipe.

Backbone	Adaptation	Dice ↑	IoU ↑
Large	Frozen	0.8747 ± 0.0025	0.7785 ± 0.0040
Large	Partial FT	0.8851 ± 0.0021	0.7953 ± 0.0031
Large	LoRA	0.8992 ± 0.0023	0.8178 ± 0.0038
Large	Full FT	$\mathbf{0.9066} \pm 0.0039$	$\mathbf{0.8301} \pm 0.0065$
Giant	Frozen	0.8753 ± 0.0036	0.7796 ± 0.0056
Giant	Partial FT	0.8911 ± 0.0015	0.8048 ± 0.0023
Giant	LoRA	$\mathbf{0.9027} \pm 0.0005$	$\mathbf{0.8234} \pm 0.0009$
Large	Frozen, BCE only	0.8810 ± 0.0018	0.7887 ± 0.0029
Large	Full FT, random initialization	0.7639 ± 0.0234	0.6426 ± 0.0240
Large	Full FT, random init., LR 10^{-4} + clip	0.7322 ± 0.0084	0.6034 ± 0.0125
nnU-Net ResEncM 3D (reference)		0.9305 ± 0.0001	0.8706 ± 0.0002

5.2. Effect of adaptation strategy

Every adaptation strategy improved mean Dice over the frozen encoder of the same backbone (Table 4, group A). On ViT-Large, Partial FT added 1.05 points, LoRA 2.46, and Full FT 3.19; on ViT-Giant, Partial FT added 1.57 and LoRA 2.73. Under the main protocol, Large Full FT reached the highest DINOv2 Dice and IoU (0.9066 ± 0.0039 and 0.8301 ± 0.0065), although without clipping its validation Dice collapsed after the selected epoch in two of three seeds (Section 5.4); Giant LoRA followed (0.9027 ± 0.0005). Their paired difference was 0.39 points ([0.10, 0.69], $p_{\text{Holm}} = 0.042$; per seed 0.20, 0.88, and 0.10; case × seed bootstrap [−0.07, 0.88], Supplementary Table S4). This gap equals the seed SD of Large Full FT and lies within the hardware retraining variation (0.38 to 0.60 points), and the main Large Full FT runs used a different software stack; it is therefore not interpreted as a strategy effect. Within ViT-Large, Full FT exceeded LoRA by 0.74 points under the main protocol ([0.40, 1.08], unadjusted; per seed 0.86, 1.00, and 0.35), larger than that variation; under the common improved recipe they did not differ detectably (Section 5.4), so this gap depended on the recipe.

Accuracy did not follow the number of trainable parameters: Large LoRA (1.25 million) exceeded Large Partial FT (51.05 million; 0.8992 versus 0.8851), and on ViT-Giant LoRA exceeded Partial FT by 1.16 points ([0.77, 1.53]).

In the two volumes shown, the median- and lowest-ranked (Figure 2), errors of the adapted models concentrate in narrow extensions of the reference mask, and coronal reformats (Supplementary Figure S4) show irregular inter-slice boundaries in the slice-wise DINOv2 predictions.

5.3. Effect of backbone scale

Replacing Large with Giant changed Dice by 0.07 points for the frozen encoder ([−0.22, 0.35]), 0.35 for LoRA ([0.16, 0.54]), and 0.59 for Partial FT ([0.30, 0.93]; Table 4, group B; Supplementary Figure S1): no detectable gain when frozen and small gains when adapted, the LoRA gain within the cross-GPU variation, and every adapted Large configuration exceeded Giant Frozen. The scale interactions (LoRA 0.28 points, [0.03, 0.55]; Partial FT 0.53, [0.11, 0.98]; p_{Holm} 0.045

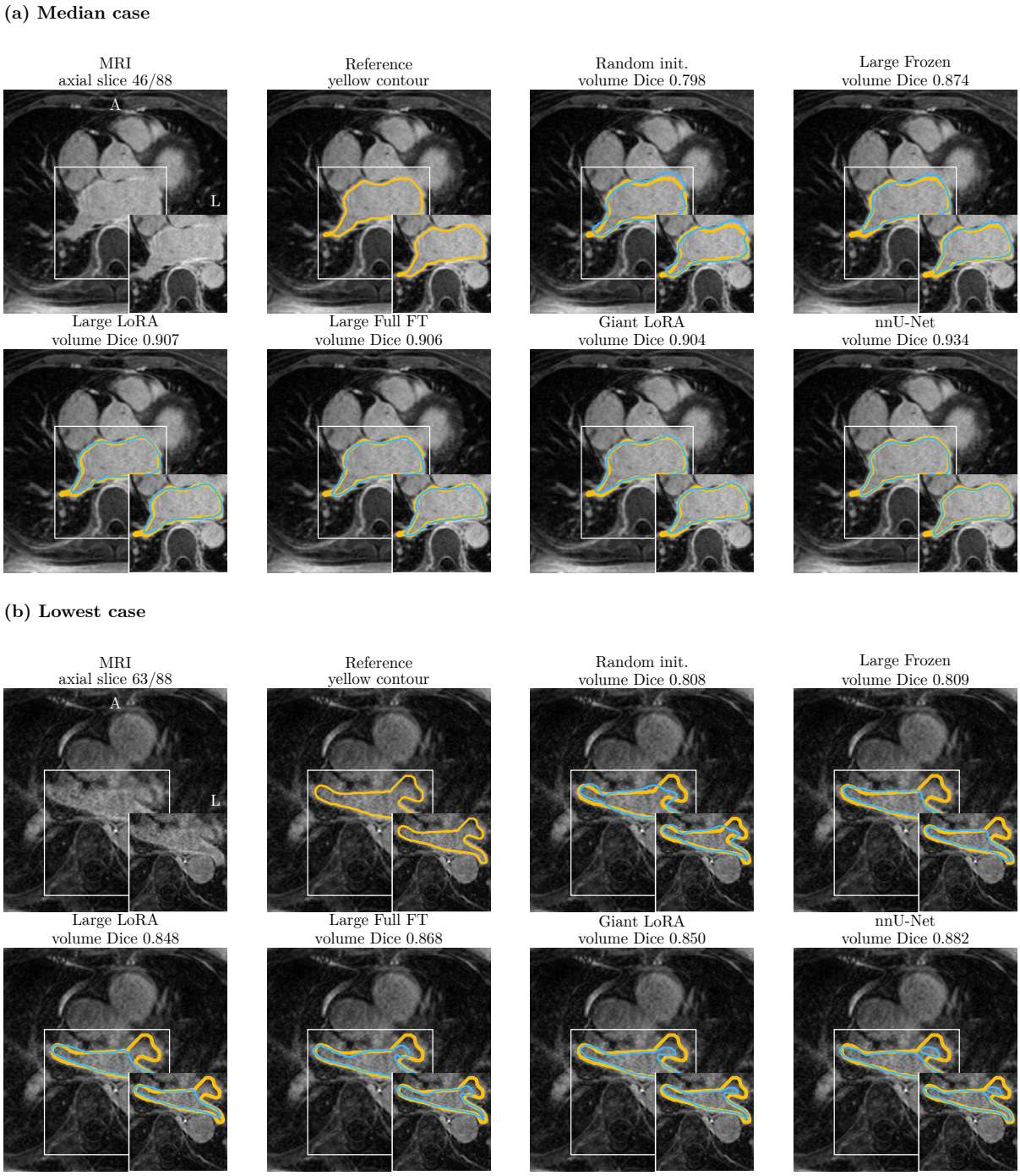


Figure 2. Predictions for (a) the median-ranked and (b) the lowest-ranked test volume, ranked by Dice averaged over the seven main-protocol DINOv2 configurations, the BCE-only frozen control, and three seeds. Each panel shows the axial slice with the largest reference LA area over a field of view that includes the heart, great vessels, and spine (A: anterior; L: patient left). Yellow (thicker, drawn below): reference; blue: prediction (seed 42); insets enlarge the boxed LA region. The image headers carry unit voxel spacing, so no physical scale bar is shown. Numbers are volume-level Dice of the displayed run, not slice Dice. Random init.: Large Full FT trained from random initialization. Main-protocol models are shown.

Table 4. Paired Dice contrasts (percentage points) for the research questions of the main protocol. For a contrast A − B, positive values indicate higher mean Dice for A; interactions are differences between two Giant − Large scale gains. Seeds are averaged within case. 95% percentile case-bootstrap CIs use 20,000 resamples (RNG seed 20260926, reset for each contrast and subset); p_{Holm} adjusts two-sided bootstrap p values over the 16 contrasts shown. Per-seed differences are given for every contrast because the case bootstrap does not include retraining variance. CIs are not Holm-adjusted, and p_{Holm} of 0.042 to 0.045 are borderline. Intervals are conditional on the observed runs and assume exchangeable test volumes. The 30-case column excludes the identified near-duplicate without retraining. Rows involving Large Full FT or Giant LoRA compare runs on different GPU models or software stacks (Supplementary Table S7); differences within the cross-GPU retraining variation (0.38 to 0.60 points) are not interpreted as effects.

Comparison	31 cases: Δ [95% CI]	p_{Holm}	Per seed	30 cases: Δ [95% CI]
A. Adaptation versus frozen extraction, same backbone				
Large: Full FT − Frozen	3.19 [2.73, 3.71]	< 0.002	2.77/3.73/3.09	3.20 [2.73, 3.73]
Large: Partial FT − Frozen	1.05 [0.81, 1.32]	< 0.002	0.67/0.99/1.48	1.07 [0.82, 1.35]
Large: LoRA − Frozen	2.46 [2.15, 2.78]	< 0.002	1.91/2.72/2.74	2.52 [2.23, 2.83]
Giant: LoRA − Frozen	2.73 [2.40, 3.14]	< 0.002	2.54/2.51/3.14	2.74 [2.39, 3.16]
Giant: Partial FT − Frozen	1.57 [1.22, 1.96]	< 0.002	1.39/1.23/2.10	1.46 [1.15, 1.76]
B. Backbone scale, Giant − Large, same strategy				
Frozen	0.07 [−0.22, 0.35]	0.626	0.02/0.34/−0.16	0.12 [−0.15, 0.40]
LoRA	0.35 [0.16, 0.54]	< 0.002	0.66/0.13/0.25	0.35 [0.16, 0.54]
Partial FT (last four blocks)	0.59 [0.30, 0.93]	< 0.002	0.75/0.57/0.46	0.51 [0.24, 0.81]
C. Scale interaction: difference between scale gains				
LoRA − Frozen	0.28 [0.03, 0.55]	0.045	0.64/−0.21/0.41	0.22 [−0.003, 0.48]
Partial FT − Frozen	0.53 [0.11, 0.98]	0.042	0.73/0.24/0.62	0.39 [0.02, 0.75]
D. Practical configuration comparisons				
Large Full FT − Giant LoRA	0.39 [0.10, 0.69]	0.042	0.20/0.88/0.10	0.34 [0.05, 0.62]
Giant LoRA − Giant Partial FT	1.16 [0.77, 1.53]	< 0.002	1.15/1.29/1.04	1.28 [1.00, 1.59]
Large LoRA − Giant Partial FT	0.81 [0.39, 1.21]	< 0.002	0.49/1.16/0.80	0.94 [0.58, 1.27]
E. Task-specific reference with a distinct recipe				
nnU-Net − Large Full FT	2.39 [1.94, 2.85]	< 0.002	2.54/1.93/2.69	2.44 [2.00, 2.90]
nnU-Net − Giant LoRA	2.78 [2.43, 3.16]	< 0.002	2.74/2.81/2.79	2.78 [2.42, 3.18]
nnU-Net − Giant Partial FT	3.94 [3.40, 4.56]	< 0.002	3.89/4.10/3.83	4.07 [3.57, 4.65]

All DINOv2 rows use the BCE–Dice objective and one loss implementation. Partial FT updates the last four blocks, not a matched fraction of depth. nnU-Net uses its distinct 3D recipe. Per-seed values follow seed order 42/2026/3407.

and 0.042) have wide intervals and are treated as exploratory. Because Giant Full FT was not evaluated, Large Full FT versus Giant LoRA changes both strategy and scale (all contrasts in Supplementary Figures S2 and S3).

5.4. Recipe sensitivity

Supplementary Table S9 reports each factor within its strategy. The four-layer readout raised Dice by 2.24 points for Large Frozen and 0.73 for Large LoRA without extra trainable parameters. The higher adapter learning rate raised Large LoRA by 0.46 points ([0.20, 0.75]) and Giant LoRA by 0.26 ([0.001, 0.51]; across GPU models), leaving a LoRA scale gain of 0.15 points ([−0.004, 0.30]). Gradient clipping prevented the validation collapse of unclipped Large Full FT, whose validation Dice had fallen to near zero after strong checkpoints in two of three seeds (MDD 0.89 and 0.89, and 0.51 in the third; Table S6), and reduced the seed SD of test Dice from 0.0039 to 0.0006.

Table 5 applies the three factors together to all four ViT-Large strategies on one hardware stack. Every strategy improved, most for the weakest (frozen encoder 2.86 points, Partial FT 2.28, LoRA 1.23, Full FT 0.49). LoRA and Full FT did not differ detectably (0.9115 ± 0.0007 and 0.9115 ± 0.0011; difference 0.00 points, [−0.19, 0.18]), although only the LoRA adapter learning rate had been swept; the gap between Full FT and the frozen encoder fell from 3.19 to 0.83

Table 5. Common improved protocol for the four ViT-Large strategies (post hoc; three seeds; all runs on one RTX 5090 and software stack). Every strategy uses the four-layer readout, gradient-norm clipping at 1.0 and the learning-rate rule (pretrained weights 10^{-5}, LoRA adapters 10^{-4}, decoder 10^{-3}); all other settings follow the main protocol, and the test set was evaluated only after all 12 runs had finished. Δ: paired case-bootstrap difference in percentage points (unadjusted 95% CI); per-seed differences follow seed order 42/2026/3407.

Strategy (ViT-Large)	Dice, main	Dice, common	Δ [95% CI]	Cases > 0	Per seed
Frozen	0.8747 ± 0.0025	0.9032 ± 0.0008	2.86 [2.44, 3.32]	31/31	2.63/2.98/2.96
Partial FT	0.8851 ± 0.0021	0.9079 ± 0.0010	2.28 [1.90, 2.73]	31/31	2.48/2.33/2.04
LoRA	0.8992 ± 0.0023	0.9115 ± 0.0007	1.23 [0.95, 1.53]	30/31	1.41/1.10/1.18
Full FT	0.9066 ± 0.0039	0.9115 ± 0.0011	0.49 [0.31, 0.68]	27/31	0.76/−0.02/0.73
Contrasts under the common improved protocol					
Full FT − LoRA			0.000 [−0.19, 0.18]	18/31	0.21/−0.11/−0.10
LoRA − Partial FT			0.36 [0.12, 0.63]	23/31	0.17/0.50/0.40
Partial FT − Frozen			0.47 [0.21, 0.75]	22/31	0.52/0.34/0.55
Full FT − Frozen			0.83 [0.55, 1.11]	28/31	0.90/0.73/0.86

Measured in these runs (one RTX 5090; Frozen, Partial FT, LoRA, Full FT): peak allocated memory 1.68, 2.89, 4.93, and 8.80 GB; epochs completed 45 / 28 / 47, 27 / 33 / 29, 34 / 26 / 35, and 26 / 15 / 15; training time excluding validation and checkpoint writing 1.11 ± 0.28, 1.05 ± 0.11, 1.57 ± 0.24, and 1.40 ± 0.49 hours (mean ± SD over seeds), or 1.7, 2.1, 3.0, and 4.5 minutes per epoch.

points ([0.55, 1.11]), and LoRA still exceeded Partial FT (0.36, [0.12, 0.63]) and Partial FT the frozen encoder (0.47, [0.21, 0.75]). Peak memory in these runs (1.7, 2.9, 4.9, and 8.8 GB for Frozen, Partial FT, LoRA, and Full FT) kept the main-protocol ordering, so LoRA reached this Dice with about 56% of the memory of Full FT. It did not train faster: its epochs were shorter (3.0 versus 4.5 minutes) but more numerous (34, 26, and 35 versus 26, 15, and 15), for 1.57 ± 0.24 versus 1.40 ± 0.49 hours of measured training time.

5.5. Reduced label budgets

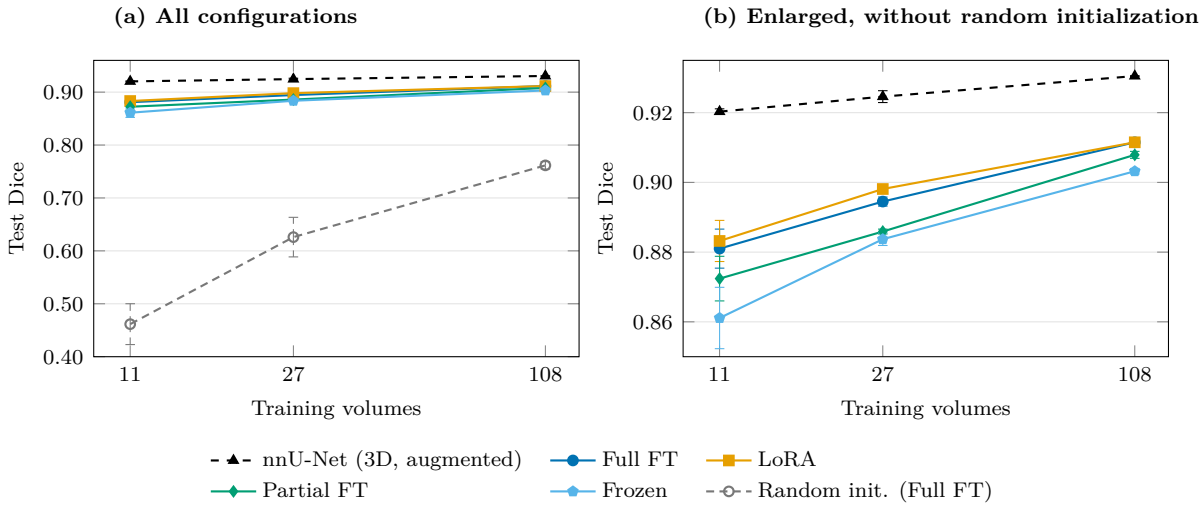


Figure 3. Test Dice with 11, 27, and 108 training volumes under the common improved recipe (ViT-Large) and for nnU-Net with its own recipe; markers are means and error bars sample SDs of the three seed means. Each seed draws one nested subset shared by all configurations, and the 15 validation and 31 test volumes are unchanged. Random init.: the same ViT-Large trained from random initialization with full fine-tuning. The horizontal axis is ordinal, not to scale. Paired contrasts and Holm-adjusted p values are in Supplementary Table S10.

Figure 3 repeats the common recipe with fewer training volumes (Supplementary Table S10). The gain of DINOv2 initialization over the randomly initialized encoder grew from 14.99 points with 108 volumes to 26.85 with 27 and 41.95 with 11 ([36.55, 47.66]; all 31 cases; change 26.96, [22.99, 31.17]), and pretrained Full FT on 11 volumes still exceeded the randomly initialized encoder on 108, by 11.94 points ([7.86, 17.26]). LoRA and Full FT did not differ detectably at either reduced budget (LoRA minus Full FT: 0.22 points with 11 volumes, [−0.17, 0.58]; 0.37 with 27, [−0.24, 1.22]), whereas the advantage of LoRA over the frozen encoder grew from 0.83 to 2.21 points (change 1.38, [0.64, 2.25]). With 11 volumes, Full FT selected its checkpoint after one or two epochs and LoRA after 9 to 23.

5.6. Augmentation of the training slices

With nnU-Net-style augmentation for both encoders (Supplementary Table S11), DINOv2 initialization exceeded random initialization by 21.46 points with 108 training volumes ([17.19, 26.49]) and 37.48 with 11 ([31.44, 44.16]), in all 31 cases ($p_{\text{Holm}} < 0.002$). Augmentation lowered the randomly initialized encoder by 6.88 points with 108 volumes ([−10.33, −3.84]), but these runs reached the 50-epoch limit with checkpoints from epochs 45 to 49 and had not converged; with 11 volumes it raised that encoder by 4.84 points ([1.18, 8.54]; per seed 0.48 to 11.30). For LoRA, augmentation changed Dice by −0.41 points with 108 volumes ([−0.74, −0.11]) and +0.15 with 11 ([−0.29, 0.56]), leaving nnU-Net 2.30 ([1.83, 2.81]) and 3.57 points ([2.87, 4.33]) ahead.

5.7. Decoder capacity

Table 6. Decoder capacity (RQ3; post hoc with respect to the main and common-recipe results, with design and planned contrasts fixed before any of these runs). ViT-Large strategies with the common improved recipe and either the lightweight decoder with the four-layer readout (0.66 million parameters; the runs of Tables 5 and S10) or a hierarchical decoder with skip connections from encoder blocks 5, 11, 17, and 23 and from the input slice (3.12 million parameters). Subsets, batch order, learning rates, checkpoint rule, and evaluation are identical, and all hierarchical-decoder runs used one RTX 5090. Dice: mean ± SD of three seed means. Δ: hierarchical minus lightweight decoder. Differences are paired case-bootstrap differences in percentage points with 95% CIs; per-seed differences follow seed order 42/2026/3407.

Strategy	Lightweight	Hierarchical	Δ [95% CI]	Per seed	nnU-Net 3D − hierarchical
108 training volumes (nnU-Net 3D: 0.9305 ± 0.0001)					
Frozen	0.9032 ± 0.0008	0.9082 ± 0.0015	0.50 [0.25, 0.75]	0.34/0.39/0.76	2.23 [1.79, 2.73]
Partial FT	0.9079 ± 0.0010	0.9092 ± 0.0033	0.13 [−0.11, 0.37]	0.25/0.39/−0.25	2.12 [1.79, 2.48]
LoRA	0.9115 ± 0.0007	0.9124 ± 0.0021	0.09 [−0.15, 0.31]	0.30/−0.19/0.16	1.81 [1.51, 2.11]
Full FT	0.9115 ± 0.0011	0.9148 ± 0.0017	0.33 [0.20, 0.46]	0.35/0.20/0.44	1.57 [1.23, 1.92]
11 training volumes (nnU-Net 3D: 0.9203 ± 0.0008)					
LoRA	0.8832 ± 0.0059	0.8764 ± 0.0171	−0.68 [−1.25, −0.19]	−2.30/0.15/0.12	4.39 [3.52, 5.43]
Full FT	0.8810 ± 0.0056	0.8535 ± 0.0083	−2.75 [−4.59, −1.38]	−2.90/−2.97/−2.37	6.68 [4.94, 9.00]
Contrasts with the hierarchical decoder, 108 training volumes (unadjusted)					
Full FT − LoRA			0.24 [0.02, 0.47]	0.26/0.29/0.18	
LoRA − Partial FT			0.32 [0.13, 0.52]	0.22/−0.08/0.82	
Partial FT − Frozen			0.10 [−0.12, 0.33]	0.43/0.33/−0.45	
Full FT − Frozen			0.66 [0.37, 0.96]	0.91/0.54/0.54	

p_{Holm} over the 12 planned contrasts: Δ for LoRA with 11 volumes 0.016, Partial FT with 108 volumes 0.560, LoRA with 108 volumes 0.560; all others < 0.002. Every planned contrast whose interval excludes zero has the same sign in all three seeds, except Δ for LoRA with 11 volumes; a case × seed bootstrap excludes zero for the other contrasts with intervals excluding zero and for no further contrast.

With 108 training volumes, the hierarchical decoder (Table 6) raised Dice by 0.50 points for the frozen encoder ([0.25, 0.75]) and 0.33 for Full FT ([0.20, 0.46]), whereas the changes for Partial

FT (0.13) and LoRA (0.09) did not exclude zero. Full FT with this decoder reached the highest DINOv2 Dice of the study (0.9148 ± 0.0017), 0.24 points above LoRA with the same decoder ([0.02, 0.47]; [−0.04, 0.52] with seeds resampled). With 11 volumes, the hierarchical decoder lowered Dice, by 2.75 points for Full FT ([−4.59, −1.38]) and 0.68 for LoRA (driven by one seed).

5.8. External evaluation

Without retraining (Table 7), the main-protocol DINOv2 configurations reached mean Dice of 0.759 to 0.779 on the 40 CDEMRIS LGE volumes and 0.528 to 0.653 on the 20 MSD bSSFP volumes, where Giant LoRA and Large Full FT were highest; on MSD, the BCE-only frozen control (0.607) exceeded the combined-loss frozen encoder (0.528) by 7.9 points, so the external ordering was sensitive even to the loss. The common-recipe models reached 0.775 to 0.784 and 0.592 to 0.637; their LoRA and Full FT models had external differences of opposite sign, both intervals including zero (Full FT minus LoRA: −0.9 points on CDEMRIS, [−2.4, 0.1]; 4.5 on MSD, [−0.02, 9.4]).

Table 7. External evaluation without retraining (RQ4, exploratory). CDEMRIS: 40 LGE volumes, one pre- and one post-ablation scan from each of 20 patients, two non-Utah centers; MSD Task02: the 20 bSSFP volumes with public reference masks (different sequence). Dice is the mean ± SD of three seed means; Δ is the paired difference configuration minus nnU-Net in percentage points (positive values favor the configuration) with an unadjusted 95% bootstrap CI that resamples patients (CDEMRIS) or volumes (MSD). Label definitions across centers may differ. The common-recipe rows were scored post hoc with the same external protocol.

Configuration	CDEMRIS Dice	Δ vs nnU-Net	MSD Dice	Δ vs nnU-Net
Large Frozen (BCE)	0.765 ± 0.004	1.0 [−2.0, 5.5]	0.607 ± 0.016	−11.9 [−14.4, −9.3]
Large Frozen	0.759 ± 0.005	0.4 [−2.8, 4.9]	0.528 ± 0.032	−19.9 [−22.9, −16.6]
Giant Frozen	0.762 ± 0.001	0.7 [−2.3, 5.2]	0.538 ± 0.018	−18.9 [−21.9, −15.6]
Large Partial FT	0.772 ± 0.004	1.7 [−1.3, 6.3]	0.553 ± 0.022	−17.4 [−23.1, −11.8]
Giant Partial FT	0.775 ± 0.004	2.0 [−0.9, 6.6]	0.560 ± 0.052	−16.6 [−20.8, −12.8]
Large LoRA	0.770 ± 0.001	1.5 [−1.4, 6.0]	0.587 ± 0.033	−14.0 [−19.1, −8.8]
Giant LoRA	0.779 ± 0.003	2.4 [−0.6, 6.9]	0.653 ± 0.027	−7.4 [−11.3, −3.0]
Large Full FT	0.776 ± 0.009	2.1 [−0.9, 6.7]	0.640 ± 0.022	−8.7 [−13.2, −3.5]
Common improved recipe (ViT-Large; post hoc)				
Frozen	0.775 ± 0.001	2.0 [−0.9, 6.5]	0.630 ± 0.028	−9.6 [−13.3, −6.0]
Partial FT	0.779 ± 0.001	2.4 [−0.7, 7.2]	0.625 ± 0.026	−10.1 [−14.0, −6.3]
LoRA	0.784 ± 0.0005	3.0 [−0.1, 7.5]	0.592 ± 0.049	−13.4 [−19.2, −8.3]
Full FT	0.775 ± 0.006	2.0 [−1.0, 6.7]	0.637 ± 0.029	−8.9 [−14.3, −4.0]
nnU-Net ResEncM 3D	0.755 ± 0.015	–	0.726 ± 0.019	–

Common recipe, Full FT − LoRA (percentage points, unadjusted 95% CI): CDEMRIS −0.9 [−2.4, 0.1], MSD 4.5 [−0.02, 9.4].

5.9. Accuracy–resource trade-offs

Parameter efficiency did not translate into proportional memory or time savings (Table 8, Figure 4). Within ViT-Large, in the fixed-workload benchmark, LoRA needed 4.9 GB of peak training memory versus 8.7 GB for Full FT and 9.6 versus 14.1 seconds per 100 batches, but it completed more epochs in each seed (25, 50, and 22 versus 18, 28, and 21; one LoRA run reached the 50-epoch limit), as under the common recipe. Giant LoRA updated about 135 times fewer parameters than Large Full FT but required 15.9 versus 8.7 GB and 26.4 versus 14.1 seconds per 100 batches.

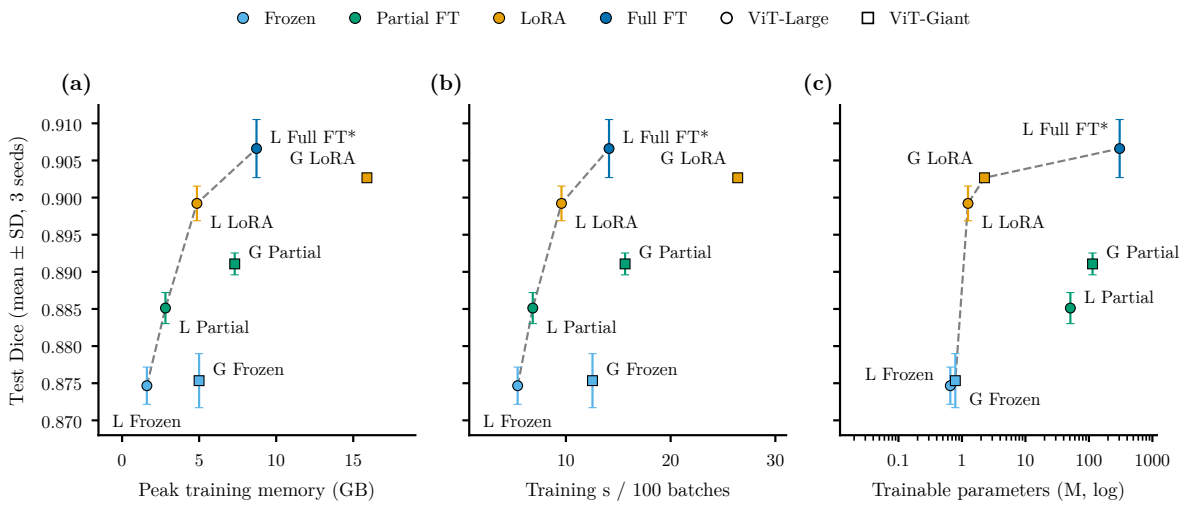


Figure 4. Observed trade-offs between accuracy and (a) peak training memory, (b) time per 100 training batches, and (c) trainable parameters under the main protocol. Dice is the three-seed mean with sample SD bars; resources come from the fixed workload (batch 4, 448 by 448, RTX 4090, three repeats of 100 batches). Dashed lines mark the observed nondominated points in each panel. Colours denote the adaptation strategy as in Figure 3; circles (L) and squares (G) denote ViT-Large and ViT-Giant; FT, fine-tuning. Large Full FT (*) had one skipped optimizer update during the benchmark. Frontiers ignore uncertainty: several Dice differences along them are within the seed SD.

Table 8. Trainable parameters, peak allocated training memory, time per 100 training batches, and completed epochs under the main protocol. Memory and time are measured in the fixed-workload benchmark on one RTX 4090 (Table S5); epochs are those completed before early stopping, in seed order 42/2026/3407. Wall-clock training times of the main-protocol runs are not compared because the runs used different GPU models.

Backbone	Adaptation	Trainable parameters	Peak memory (GB)	Seconds per 100 batches	Epochs
Large	Frozen	655,329	1.609	5.4	31 / 26 / 34
Large	Partial FT	51,048,417	2.814	6.8	25 / 32 / 16
Large	LoRA	1,245,153	4.854	9.6	25 / 50 / 22
Large	Full FT	305,023,969	8.722	14.1	18 / 28 / 21
Giant	Frozen	786,401	4.997	12.5	19 / 21 / 31
Giant	Partial FT	114,132,961	7.309	15.7	43 / 42 / 49
Giant	LoRA	2,260,961	15.888	26.4	32 / 36 / 27

The BCE-only frozen control has the Large Frozen architecture (655,329 trainable parameters, 1.609 GB) and completed 21 / 26 / 28 epochs.

5.10. Comparison with a strong task-specific reference

The nnU-Net ResEncM 3D reference reached 0.9305 ± 0.0001, 2.39 points above Large Full FT ([1.94, 2.85]) and 2.78 above Giant LoRA ([2.43, 3.16]), a pipeline-level comparison (Section 4.3), and in the lowest-ranked volume shown it recovered narrow extensions of the reference mask more completely (Figure 2). It remained above every DINOv2 configuration of the study, by 1.90 points over the common-recipe Full FT and LoRA and 1.57 over Full FT with the hierarchical decoder ([1.23, 1.92]). The 2D configuration with the same recipe reached 0.9233 ± 0.0005, 0.72 points below the 3D reference ([0.40, 1.06]) but 0.84 points above the best DINOv2 configuration ([0.52, 1.17]; Supplementary Table S8).

With fewer training volumes the gap widened (Figure 3): nnU-Net lost only 1.02 points ([0.65, 1.44]) between 108 and 11 volumes, so its advantage over LoRA grew from 1.90 to 3.71 points ([3.05, 4.42]; change 1.82, [1.30, 2.39]; all 31 cases at 11 volumes), and over the hierarchical-decoder LoRA and Full FT to 4.39 and 6.68 points. On 11 volumes, nnU-Net even exceeded LoRA and Full FT on all 108, by 0.88 points ([0.27, 1.43] and [0.26, 1.44]; exploratory).

Externally, this advantage was not observed on LGE data from other centers: every DINOv2 configuration had higher mean Dice than nnU-Net on CDEMRIS (0.755), but no paired difference excluded zero and median per-case differences ranged from -1.7 to $+0.4$ points, so neither a difference nor equivalence was shown. The means were dominated by one volume in which only nnU-Net failed, for all seeds (Dice 0.02, versus 0.78 to 0.87 for DINOv2); without it (post hoc), the differences ranged from -1.5 to $+0.9$ points, all intervals including zero. A second volume was missed by every model (Dice at most 0.19); without both (post hoc), the differences ranged from -2.1 to $+1.0$ points, and only Large Frozen differed, 2.06 points below nnU-Net ([-3.72, -0.54]). On the MSD bSSFP data, every DINOv2 configuration was lower than nnU-Net (0.726), by 7.4 points ([3.0, 11.3]) for Giant LoRA to 19.9 points for Large Frozen, and the common-recipe models by 8.9 to 13.4 points.

6. Discussion

6.1. What the controlled comparison shows

Under one protocol and within the evaluated ranges, DINOv2 initialization raised Dice by 14.3 points over an unaugmented ViT trained from scratch, by 14.99 to 41.95 under the common recipe as the training set shrank, and by 21.46 to 37.48 when both encoders were augmented; encoder adaptation and decoder readout changed Dice by about 1 to 3 points, backbone scale by at most 0.6, and decoder capacity by at most 0.5 with all training volumes but by -2.75 with 11. Because the augmented from-scratch runs with 108 volumes had not converged within 50 epochs, these magnitudes describe the levels compared rather than a ranking of factors. Most strategy gaps of the main protocol depended on the recipe: with a common readout, clipping, and learning-rate rule, LoRA and full fine-tuning did not differ detectably and the frozen encoder came within 0.83 points (2.2 behind LoRA with 11 volumes). Method rankings depend on evaluation choices [17] and here also on training choices, much as parameter-efficient fine-tuning depended on backbone size in a SAM benchmark [13]; on chest-radiograph foundation models, LoRA exceeded full fine-tuning in 13 of 18 transfer tasks [38].

LoRA outperformed partial fine-tuning under both protocols despite updating about 41 times fewer parameters on ViT-Large, possibly because low-rank updates in all blocks permit adjustments unavailable when only the final blocks change, which we did not test. Much of the main-protocol gap between frozen and adapted encoders came from the last-layer readout, in line with gains from multi-level features and stronger decoders in other DINOv2 studies [6, 7, 10, 12] and a broader benchmark [15]; once the readout drew on several blocks, a hierarchical decoder added at most 0.5 points.

Because the scale gains were small, recipe-dependent, and partly within the cross-GPU variation, a larger encoder offered little return at its higher cost, as larger SAM backbones did elsewhere [13]. Unstable gradients have been linked to degraded full fine-tuning with a newly initialized decoder [15], consistent with the collapse that clipping prevented here.

The study proposes no new network; its contribution is the comparison itself, which adds to the prior DINOv2 LA studies the four adaptation strategies under one protocol, a no-pretraining control with and without augmentation, measured training resources, and reduced label budgets on one split. Two observations may transfer to other ViT-based segmentation tasks: strategy rankings can change with the training recipe, and trainable-parameter counts do not determine training memory or time. The absolute Dice values and the size of the pretraining gain are specific to this dataset and protocol.

6.2. Relation to prior work and to a strong task-specific reference

These findings complement prior DINOv2-based LA work [9, 10, 12] by separating adaptation strategy, backbone scale, and pretraining under a common decoder topology. Kundu et al. reported DINOv2 with U-Net-style decoders above a 2D nnU-Net, including on the 2018 challenge data [10], whereas here every DINOv2 configuration remained below the reference internally (Section 5.10).

Three differences may contribute. First, the reference: we used the residual-encoder 3D configuration, recommended after many reported gains over nnU-Net did not hold under rigorous validation [5], whereas the prior LA studies used a 2D nnU-Net [10, 12]; the 2D configuration here also exceeded every DINOv2 model, so dimensionality alone does not explain the gap. Second, the decoder: prior decoders use skip connections, multi-scale fusion, or image input, but the hierarchical decoder here narrowed the gap only from 1.90 to 1.57 points. Third, the pipeline: nnU-Net adds augmentation, deep supervision, and a 1,000-epoch schedule; augmentation helps most for very small training sets in other tasks [39], but adding it to LoRA left nnU-Net 2.30 and 3.57 points ahead. Partitions and preprocessing also differ, so the gap cannot be assigned to one component. Training and scoring a strong reference on the same cases therefore places DINOv2 adaptation relative to an established pipeline under identical evaluation.

Relative to random initialization, natural-image pretraining reduced the annotation needed, as anticipated [7, 9]: pretrained Full FT on 11 volumes outperformed the randomly initialized encoder on 108. nnU-Net was nevertheless best at every budget and, with 11 volumes, exceeded every DINOv2 configuration trained on 108; dimensionality, deep supervision, or the longer schedule remain candidate reasons. Fine-tuned SAM and MedSAM likewise did not outperform a standard U-Net on brain MRI with very few training volumes [40], and a task-specific 3D LA network reached 0.921 Dice with 16 training scans of the 2018 challenge data [19].

Externally, the DINOv2 models reached mean Dice of 0.759 to 0.784 on LGE data from other centers, against 0.755 for nnU-Net, with every paired interval including zero, whereas on bSSFP data they were clearly lower, for reasons not isolated here. The common improved recipe changed neither finding, and neither supports a general robustness advantage of the pretrained encoder.

6.3. Practical recommendations

Table 9 turns the evidence into a guide for practitioners who build an LA segmentation model on a DINOv2 encoder: a starting configuration, alternatives by training memory, components with little gain in the tested settings, and when a task-specific pipeline is preferable. It applies to the evaluated protocol and data size and does not replace validation on a new dataset.

Table 9. Practical guide for building an LA segmentation model on a DINOv2 encoder. Effects are paired differences in Dice percentage points on the internal test set unless stated otherwise; memory and time are measured. Much of the evidence comes from post hoc analyses (the common improved recipe, label budgets, and decoder study, all on ViT-Large), so recommendations resting on them are provisional.

Recommendation	Supporting evidence and boundary
Always. Initialize the ViT encoder from DINOv2 rather than training it from scratch.	+14.3 points over an unaugmented ViT from scratch, all 31 cases; a second from-scratch recipe (higher learning rate, clipping) was lower. Under the common recipe the gain was +14.99 points with 108 training volumes and +41.95 with 11, and +21.46 and +37.48 with nnU-Net-style augmentation for both encoders (all 31 cases). The augmented from-scratch runs with 108 volumes had not converged within 50 epochs.
Starting point, especially with limited memory. ViT-Large with LoRA (rank 4), a four-layer readout, gradient clipping, and separate learning rates (10^{-5} pretrained weights, 10^{-4} adapters, 10^{-3} decoder).	No detectable difference from Large Full FT under this recipe (0.9115 each; [−0.19, 0.18]) with about 56% of its peak training memory (4.9 versus 8.8 GB), but not less training time (1.57 versus 1.40 hours); no detectable difference with 11 or 27 training volumes (LoRA minus Full FT: 0.22, [−0.17, 0.58], and 0.37, [−0.24, 1.22]). Only the LoRA adapter learning rate was swept (rank 4); Full FT kept the shared learning rates.
More memory. Full fine-tuning, always with gradient clipping.	Clipping prevented validation collapse in all seeds and raised Large Full FT by 0.59 points (on a different GPU model from the unclipped runs). With the common recipe Full FT did not differ detectably from LoRA, and with the hierarchical decoder it was 0.24 points higher ([0.02, 0.47]; [−0.04, 0.52] with seeds resampled); externally, the difference changed sign between datasets, with intervals including zero.
Less memory. Partial FT, or a frozen encoder with the four-layer readout unless annotated volumes are very few.	With the common recipe, Partial FT was 0.36 points below LoRA ([0.12, 0.63]) at 2.9 GB, and the frozen encoder 0.83 points below Full FT at 1.7 GB but 2.21 points below LoRA with 11 training volumes. The readout alone added 2.24 (Frozen) and 0.73 (LoRA) points without extra trainable parameters.
Little gain in the tested settings. A larger backbone (ViT-Giant).	Giant − Large: 0.07 to 0.59 points under the main protocol, 0.15 for LoRA at the higher adapter rate, partly within the cross-GPU variation; Giant was not run with the common recipe; 2.6 to 3.3 times the training memory of the matched Large configuration. Giant LoRA had about 135 times fewer trainable parameters than Large Full FT but needed more memory (15.9 versus 8.7 GB) and time per batch, so few trainable parameters do not imply low compute.
Little gain in the tested settings. A larger decoder with skip connections.	One hierarchical decoder design (3.12 versus 0.66 million parameters) added 0.09 to 0.50 points with 108 training volumes and lowered Dice with 11 (by 2.75 points for Full FT).
Little gain in the tested settings. nnU-Net-style augmentation of the training slices with LoRA.	−0.41 points with 108 training volumes ([−0.74, −0.11]) and +0.15 with 11 ([−0.29, 0.56]); it did not close the gap to nnU-Net.
Reporting. Report the training recipe together with strategy rankings.	The common improved recipe changed the Full FT−Frozen gap from 3.19 to 0.83 points and the Full FT−LoRA gap from 0.74 to 0.00; single-factor changes moved the Full FT–Giant LoRA gap between 0.13 and 0.98 points.
Alternative. If a ViT encoder is not required and the highest in-domain Dice is the goal, prefer a task-specific nnU-Net, especially with few annotated volumes.	Pipeline-level comparison (3D, augmentation, own schedule), so the source of the gap was not isolated: +2.39 points over Large Full FT internally (1.90 over the common-recipe Full FT and LoRA, 1.57 over Full FT with the hierarchical decoder), widening to 3.71 over LoRA with 11 training volumes, and 2.30 and 3.57 over LoRA trained with nnU-Net-style augmentation; on external LGE data 0.755 versus 0.759 to 0.784 for the DINOv2 models, all paired intervals including zero; higher on bSSFP data.

6.4. Limitations

Because the public collection provides no reliable patient mapping, the split is by volume, and repeat scans of one patient in both training and test data cannot be excluded; internal Dice may therefore be optimistic, although excluding the near-duplicate test volume lowered the mean Dice of the pretrained configurations and nnU-Net by at most 0.24 points (0.57 for the random-initialization control) without changing their ordering (Supplementary Tables S3 and S4). The external evaluation is inference-only, small, and may involve different label definitions.

All recipe, label-budget, decoder, and augmentation analyses were post hoc and scored the same 31 test volumes (Supplementary Table S12), so their estimates may be optimistic. The common improved recipe is one recipe rather than a per-strategy optimization (only the adapter learning rate was swept) and covers ViT-Large only, so the scale comparison rests mainly on the main protocol, and the label-budget, decoder, and augmentation models were not evaluated externally. The reduced budgets were drawn once per seed, so subset and retraining variation are only jointly resampled, and all budgets kept the 15 annotated validation volumes and the nnU-Net plans from all 123 development volumes, so the annotation used exceeds the training count.

The conclusions are further restricted to one split, three seeds (one Large LoRA run and two runs of the second from-scratch recipe reached the 50-epoch limit), a 2D formulation without inter-slice modeling, one hierarchical decoder design, augmentation in one post hoc analysis with a 50-epoch limit, LoRA rank 4, a four-block partial depth, a decoder input width that differs between backbones, and natural-image pretraining, whereas DINO pretraining on cardiac MR images has improved over natural-image foundation models across cardiac MRI tasks [41]. Only overlap metrics are reported, because the image headers carry unit voxel spacing, and Dice differences are not related to inter-observer variability. No claims are made about scar delineation, clinical utility, or patient outcomes.

7. Conclusion

AdaDINO-LA characterizes DINOv2 adaptation for LA segmentation from LGE-MRI under one specified protocol. Against the same ViT trained from scratch, DINOv2 initialization raised Dice by 14 to 21 points with all training volumes and by 37 to 42 with 11, with or without augmentation. Under the main protocol, Large Full FT gave the highest DINOv2 overlap; with one improved recipe applied post hoc to all ViT-Large strategies, LoRA and full fine-tuning did not differ detectably, with LoRA using about 56% of the peak training memory. A larger encoder and a hierarchical decoder added little in the tested settings, whereas the adapter learning rate, gradient clipping, and especially the decoder readout changed the gaps between strategies, so such comparisons should be reported with their recipe. A strong task-specific 3D nnU-Net reference remained more accurate internally, more so with fewer training volumes, also against LoRA trained with nnU-Net-style augmentation, and on bSSFP data. For practitioners building on DINOv2, these results support ViT-Large with LoRA, a four-layer readout, gradient clipping, and separate learning rates as a provisional starting point (Table 9).

CRediT authorship contribution statement

Quoc Khanh Be: Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft. **Nam Hai Tran:** Investigation, Writing – review & editing. **Gia Duc Trung Nguyen:** Investigation, Writing – review & editing. **Quynh Dao Thi Thuy:** Conceptualization, Supervision, Project administration, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics statement

This study used only publicly available, de-identified imaging data (the 2018 Left Atrial Segmentation Challenge collection, CDEMRIS, and the Medical Segmentation Decathlon) under the providers' terms of use. Ethical approval and informed consent were obtained by the original data providers; no additional approval was required for this secondary analysis.

Data availability

The Utah2018 LGE-MRI data are available from the original challenge provider under its access conditions and are not redistributed by the authors. Source code, configuration files, case lists, per-case metrics, and the scripts that regenerate the reported tables and quantitative figures are publicly available at `https://github.com/qkhanhbe/AdaDINO-LA`. Trained model weights are available from the corresponding author upon reasonable request and will be deposited in a public archive upon publication.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Codex (OpenAI) and Claude (Anthropic) to assist with programming the data-preparation, experiment-management, evaluation, and analysis scripts and with drafting and language editing of the manuscript; they were not used to create or modify images or labels. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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